

SUJAY SHIVA PRAKASH

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EDUCATION

M.S, Data Science, Analytics and Engineering (Computing and Decision Analytics) Graduating Dec 2026
Arizona State University, Tempe, AZ
Relevant coursework: Statistics for Data Analytics, Statistical Machine Learning, Data Processing at Scale, Data Visualization, Artificial Intelligence, Data Structures and Algorithms, Data Mining, Cyber Security, Optimization.

Bachelor of Engineering in Computer Science and Engineering August 2018 - June 2022
Visvesvaraya Technological University
Relevant coursework: DBMS, Design and Analysis of Algorithms, Big Data Analytics, Network Management

TECHNICAL SKILLS

Generative AI and Agents: LLM Integration, RAG Pipeline Development, LangChain, LangGraph, AI Agent Design, FAISS, Hugging Face, RAGAS
Distributed Data Engineering: Python, PySpark, Apache Spark, Airflow, Parquet, Feature Engineering, Large-Scale Data Processing, Git
Observability and Production AI: Agent Evaluation, Production Monitoring, Structured Logging, Latency Tracking, Documentation

PROFESSIONAL EXPERIENCE

Arizona Water Innovation Initiative, ASU, Tempe, AZ: Graduate Research Assistant 2025 – Present
• Led data collection and analysis for ADOT’s urban freeway water efficiency project, capturing irrigation performance, plant inventories, and water application metrics across 250+ miles of Phoenix freeways. Integrated plant health, soil moisture, weather, and land orientation data to identify overuse patterns; delivered visualizations and recommendations that informed ADOT’s irrigation design standards.
• Analyzed plant survival by vegetation type, species, and site; benchmarked actual water usage against design specifications and compared West vs. East zone diversity to surface irrigation inefficiencies.

Open Systems International, Bengaluru: Research & Development Intern Mar 2023 – Feb 2024
• Deployed automated data-driven QA workflows, reducing manual testing by 30% and enhancing production system reliability.
• Built Python pipelines to ingest and preprocess large-scale operational datasets for downstream analytics and reporting.
• Optimized data storage and querying using MongoDB and SQL to accelerate root-cause analysis and performance insights.
• Managed data quality, monitoring, and stakeholder requirements to deliver scalable data processes for critical workflows.
• Designed and maintained Jenkins/Docker CI/CD pipelines across multi-environment deployments, reducing release friction and improving rollout reliability through automated build monitoring, runbooks, and release documentation; developed TestComplete and Selenium regression suites, enforced code review standards, and tracked defect resolution to achieve zero critical escapes to production.

PROJECTS

Banking Financial Intelligence Agent — | Python | PySpark | LangChain | FAISS | OpenAI API | RAGAS | Hugging Face | Jan 2025 – May 2025
Multi-agent RAG system over financial documents backed by a PySpark large-scale transaction pipeline with agent evaluation and observability.
• Built a LangChain multi-agent orchestration layer decomposing complex banking queries into retrieval, reasoning, and structured response subtasks over 50,000 plus SEC and financial regulatory document chunks indexed in a FAISS vector store using Hugging Face embeddings.
• Developed a PySpark distributed preprocessing pipeline ingesting and transforming 590,000 plus transaction records from the IEEE-CIS fraud dataset to generate structured context injected into agent retrieval workflows. Implemented output guardrails filtering hallucinated, off-topic, and sensitive content from LLM responses, and instrumented the full agent pipeline with structured observability logging tracking retrieval precision, latency, and failure rates.

IBM BOM Hackathon — | Python | REST API | Git | Cross-functional Collaboration | Agile | April 2026 - May 2026
Built a Bill of Materials data product from the ground up with eammates in a founder-style hackathon environment under tight delivery timelines.
• Collaborated with a cross-functional team of engineers, analysts, and domain experts across multiple countries to build an IBM BOM data product from scratch within a compressed hackathon timeline, wearing multiple hats across data engineering, feature development, and stakeholder communication.
• Bridged the gap between user needs and technical constraints by translating business requirements from non-technical country stakeholders into product specifications, ensuring every shipped feature delivered measurable value aligned with team priorities.
• Contributed shipping Python code within the first day of the hackathon, applying object-oriented design patterns and participating in peer code reviews to maintain code quality and consistency across a distributed team working in parallel.
• Communicated development status and milestone progress to cross-functional stakeholders throughout the build, documenting product objectives and constraints that guided team prioritization decisions from ideation through final delivery.

RESEARCH

Quantifying Cognitive Deterioration Trajectories in Alzheimer’s Disease Using Functional Time Series Analysis: A Knowledge Graph-Driven Framework for Cross-Modal Neurodegenerative Calibration 2025 – Present
National Alzheimer’s Coordinating Center(NACC) Clinical Dataset
• Engineered a multi-stage Alzheimer’s progression pipeline on 18,516 patients (125K+ NACC records), extracting delta biomarkers (CSF tau, amyloid PET, DATSCAN) and deploying XGBoost (AUC=0.77, F1=0.77) to classify cognitive deterioration, revealing that longitudinal biomarker trajectories outperform single time-point snapshots as predictive signals.
• Designed an age-stratified Functional PCA framework treating longitudinal MMSE scores as continuous curves, decomposing individual trajectories into clinically interpretable weights, w1 capturing decline rate (63–71% variance) and w2 capturing baseline cognitive level (14–20% variance) across four age cohorts, eliminating the global-model assumption ignored by prior literature.
• Built a cumulative Random Forest prediction system growing from 104 to 330 features across 6 annual visits, achieving w1 R²=0.91 and w2 R²=0.82 by Year 5 with CV–Test gap below 0.04; pinpointed visuospatial ability (PENTAGON) and family history as dominant trajectory shape predictors, and vascular health as the primary determinant of baseline cognitive level, providing actionable targets for early clinical intervention.